I Evidence stream

unit under a known condition z



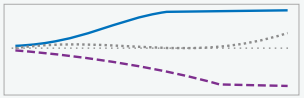
evidence $x(t)$ on $t_0 < \ldots < t_N$

history $\mathcal{F}_t = \{x(t_0{:}t), z\}$
late outcome $y \in \{0, 1\}$

delay cost λ per unit time
error costs c_{FP}, c_{FN}

II Posterior process

$\hat{p}_t \approx \Pr(y{=}1 \mid \mathcal{F}_t)$
logistic or prefix GRU



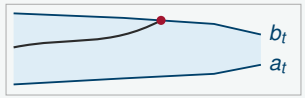
posterior paths; dotted: $p^\star$

calibration error at the stopping time:
$\kappa(\tau) \approx \mathrm{ECE}_{\mathrm{stop}}$

self-sufficiency: history adds nothing beyond $\hat{p}_t$

III Generate (Thm. 1)

$\min_\pi \mathbb{E}[\lambda\tau]$ subject to
$\mathrm{FP} \le \alpha_{\mathrm{FP}}$, $\mathrm{FN} \le \alpha_{\mathrm{FN}}$



continue in $(a_t, b_t) \ni p^\star$; • stop

backward induction per μ
$\Rightarrow$ rule family $\mathcal{G} = \{\pi_\mu\}$

fit-free special case:
margin $|\hat{p}_t - p^\star| \ge m$

IV Select & certify

select the fastest feasible rule, else the endpoint assay

units independent
Thm. 3: slack U_n, w.p. $1-\delta$

new conditions
Thm. 4: risk $\le \alpha$ in expectation

m=9 conditions: Thm. 3 slack is vacuous, use Thm. 4

deploy: stop if $\hat{p}_t \notin (a_t, b_t)$

Thm. 2 excess realised cost of a deployed rule over the best rule with the same information $\le 2L\bar{\kappa} + \eta$
(η: discretisation slack); the information deficit $V_q - V$ is unchanged by recalibration.
Cor. 1 |believed − realised cost| $\le L\,\kappa(\tau)$ for every rule, so early-decision rules must be scored on realised cost.